

15 Halogen compounds

AS · 2026 examination · Concept guide and original exam-style practice

Understand the concept

Haloalkanes undergo nucleophilic substitution when a nucleophile attacks δ^+ carbon; aqueous hydroxide produces an alcohol, while ethanolic hydroxide favours elimination to alkene. Rate depends on bond strength and structure. Halogenoarenes resist substitution under mild conditions due to resonance affecting C-halogen bond.

Worked example: follow the reasoning

Bromoethane + $\text{OH}^-(\text{aq}) \rightarrow$ ethanol + Br^- ; warming under reflux improves conversion without losing volatiles.

Quick check · multiple choice

A Hot ethanolic OH^- typically promotes which haloalkane reaction? [1]

- A Oxidation
- B Polymerisation
- C Elimination
- D Ester hydrolysis

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

1 Predict organic product when 2-bromopropane reacts with hot ethanolic KOH and explain the role of base. [3]

Answers and explanation

Quick check: C. Elimination

Question 1 [3 marks]

Propene [1]; OH^- removes β -hydrogen [1], and HBr is eliminated to make $\text{C}=\text{C}$ [1].

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or calculation error; then retry without looking at the solution.

Syllabus: Chemistry 9701, 2025–2027. <https://www.cambridgeinternational.org/Images/664563-2025-2027-syllabus.pdf>

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